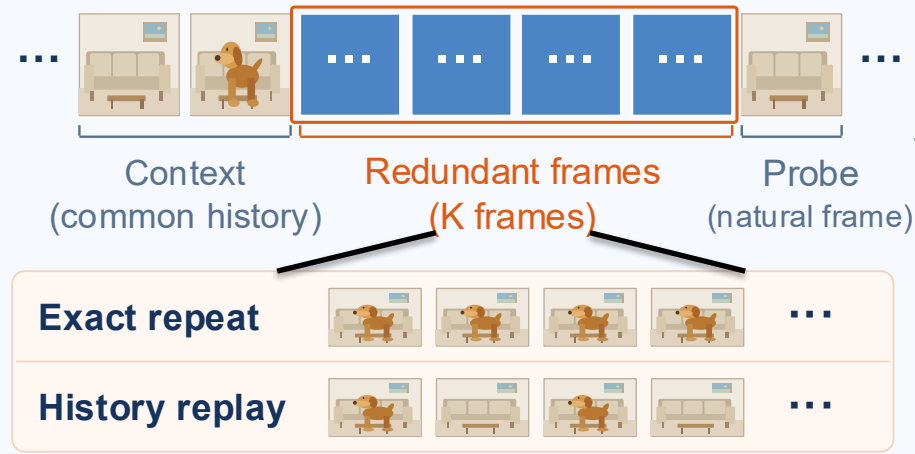
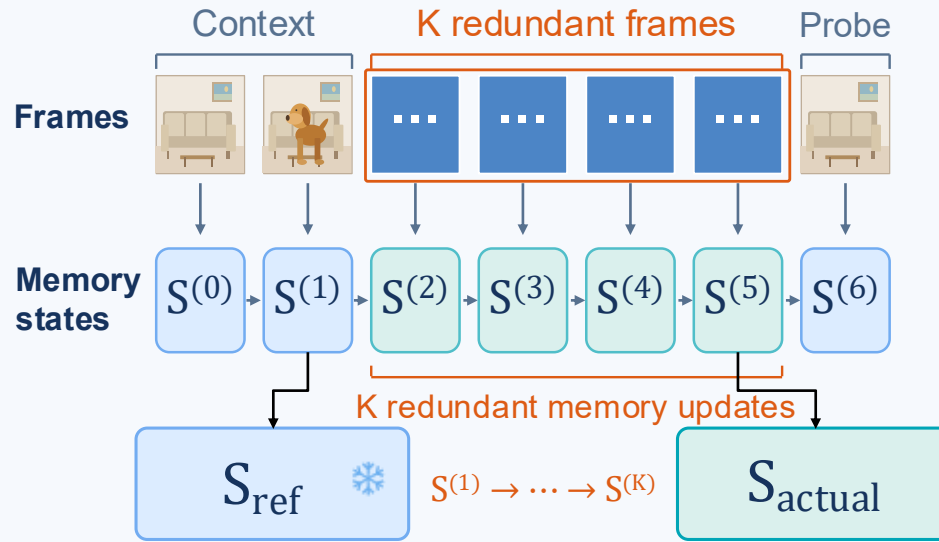


RA Redundancy-aware memory supervision

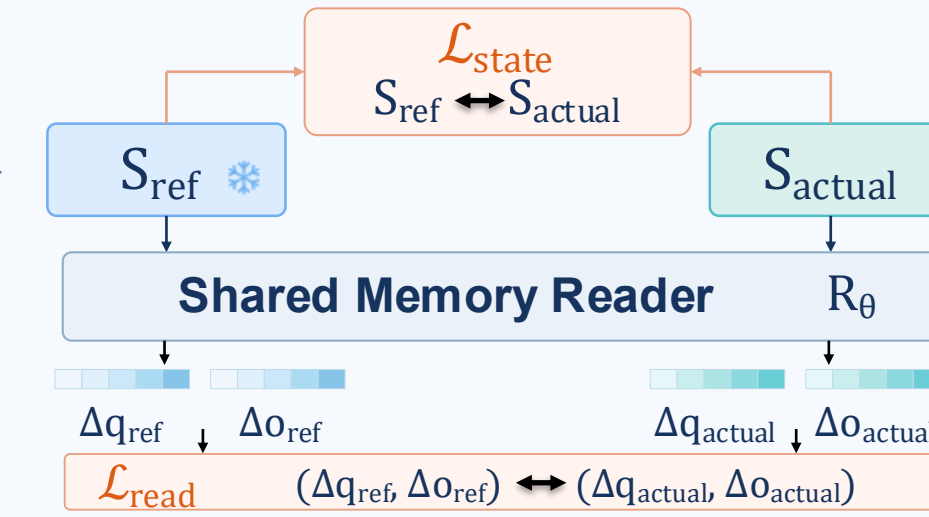
1 Redundant training pairs



2 Memory states



3 Supervise state and read consistency



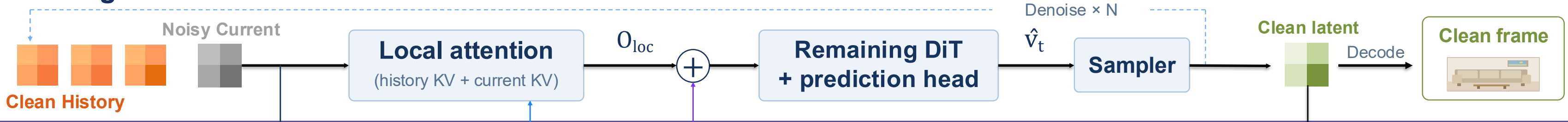
FM Generation supervision

$$\mathcal{L}_{FM} = \mathbb{E} \|\hat{v}_\theta - v\|^2$$

Training objective

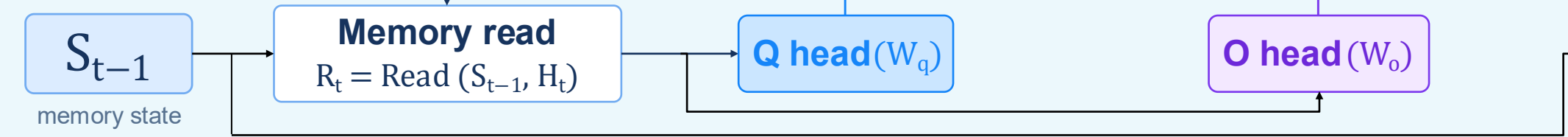
$$\mathcal{L} = \mathcal{L}_{FM} + \lambda_{RA}(\mathcal{L}_{state} + \mathcal{L}_{read})$$

Denoising

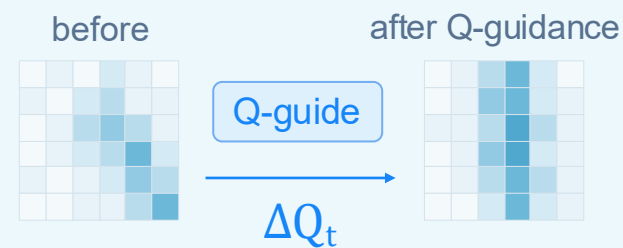


Delta Video (Ours)

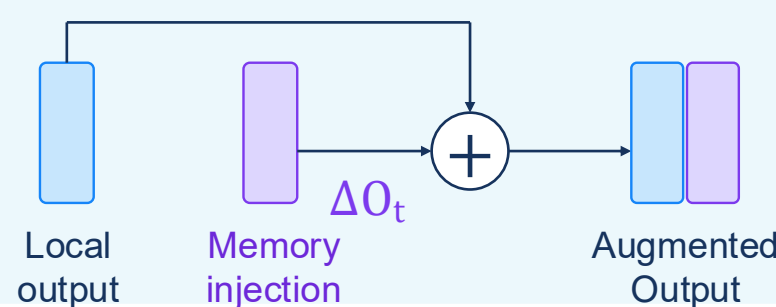
Read & inject



Q: guide local attention



O: inject memory features



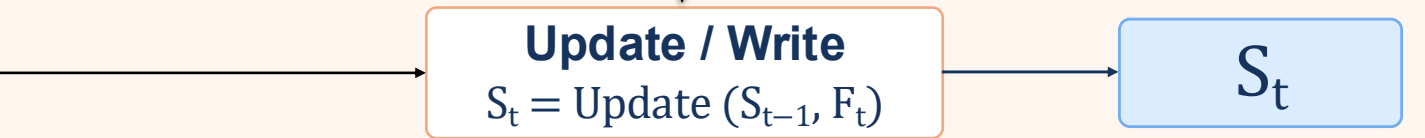
$$\Delta Q_t = W_q R_t$$

Modulate attention selection

$$\Delta O_t = W_o R_t$$

Supply historical content

Write & update



Frame-wise memory write

